

MILA ACTIVITY

TOPIC : SERVER VISUALIZATION

PRESENTED BY

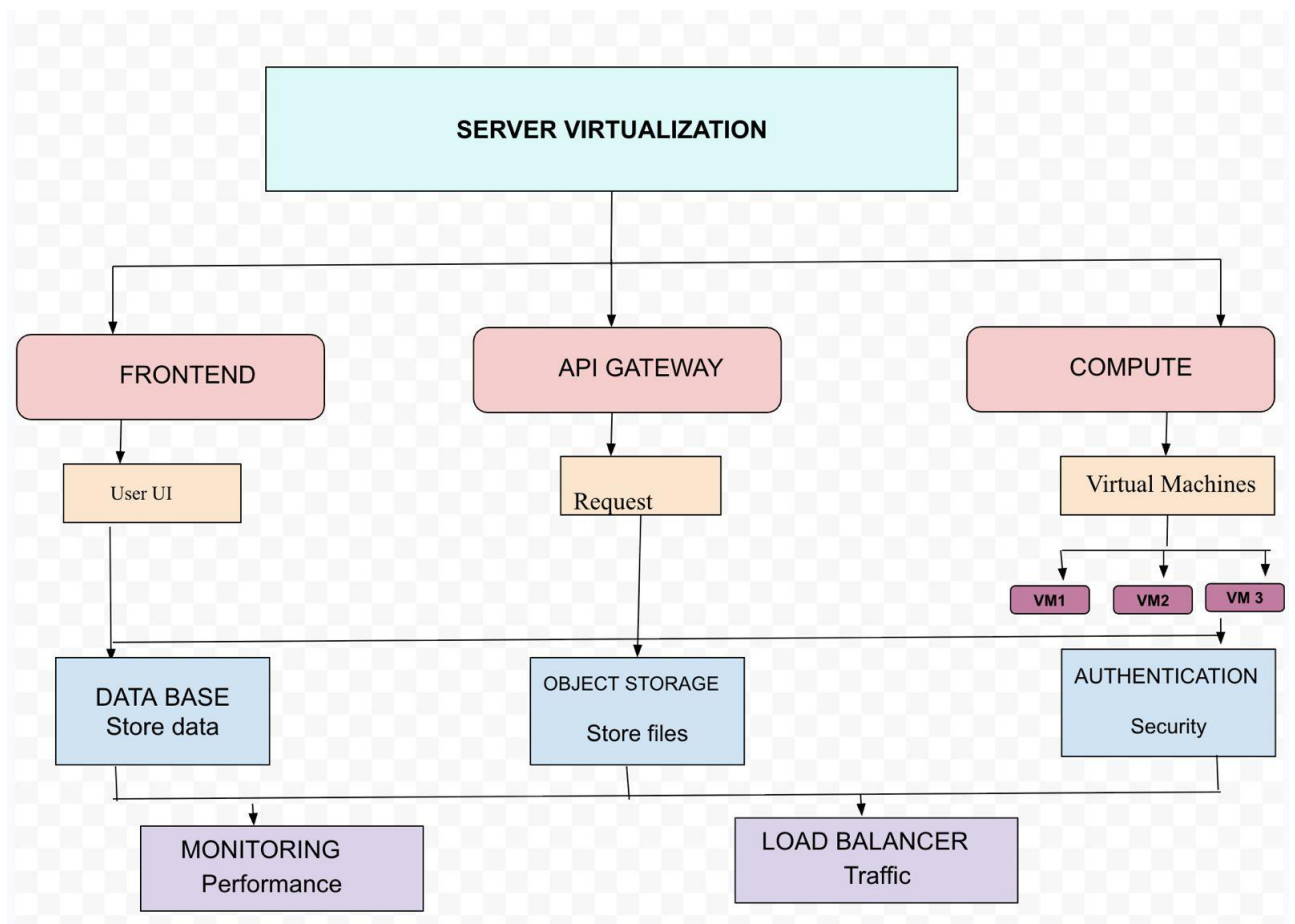
MADHU BAVANA.B (192524103)

PREETE.V(192572004)

NIVASINI V M (192511356)

KEERTHANA P (192521289)

DEVI SRILATHA.M(192525236)



Main Topic: Server Virtualization

Server virtualization means dividing one physical server into multiple Virtual Machines (VMs) using a hypervisor. Each VM can work like an independent server.

1. Frontend → User Interface

- The Frontend is the part that users can see and interact with.
- Provides a dashboard.
- Allows users to create and manage VMs.
- Shows server and VM information.
- Example: A cloud dashboard used to create a VM.

2. API Gateway → Request Routing

- The API Gateway connects the frontend with backend services.
- Receives user requests.
- Sends requests to the correct service.
- Controls and manages API access.
- Example: A request to create a VM is sent to the Compute service.

3. Compute → Virtual Machines

- Compute provides the processing resources required to run applications and VMs.
- Provides CPU and RAM.
- Creates and manages VMs.
- Uses a hypervisor to divide physical resources.

Physical Server



Hypervisor



VM1 VM2 VM3

4. Database → Store Data

- The Database stores organized information required by the system.
- User information
- VM configuration
- Application data
- Resource information
- Example: Storing the CPU and RAM configuration of a VM.

5. Object Storage → Store Files

- Object Storage stores large files and objects.
- VM images
- ISO files
- Backups
- Documents
- Example: Storing an Ubuntu ISO file for installing an operating system in a VM.

6. Authentication → Security

- Authentication verifies the identity of users.
- Login
- User verification
- Access control
- Authorization
- Example: Only an authorized user can create or delete a VM.

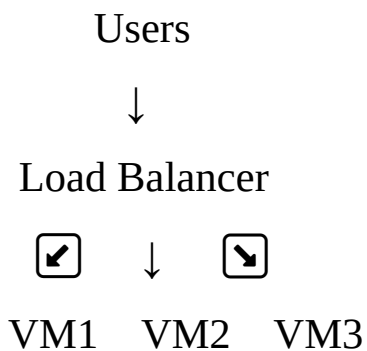
7. Monitoring → Performance Checking

- Monitoring checks the health and performance of the virtualized servers.
- It monitors:
- CPU usage
- RAM usage

- Storage
- Network
- VM status
- Example: Giving an alert when CPU usage becomes too high.

8. Load Balancer → Traffic Distribution

A Load Balancer distributes incoming requests among multiple VMs or servers.



This prevents one VM from becoming overloaded.

